

8. Generality of Data Distribution Lens

To probe the generality of the data distribution length, we design experiments using LLMs with various architectures, sizes, and temperatures.

8.1. Internal Validity

Experiment setup. For rigor, we conduct the experiments of task, length, and format generalization by training LLMs with GPT and LLaMA architectures and sizes ranging from 62K to 3B.

Findings. As illustrated in Figure 8, CoT reasoning produced by LLM with different sizes and architectures behaves similarly when encountering distribution shifts on task, length, and format generalization, highlighting the good internal validity. We further study the effect of temperature and the role of SFT with different model sizes in Appendix F.4.

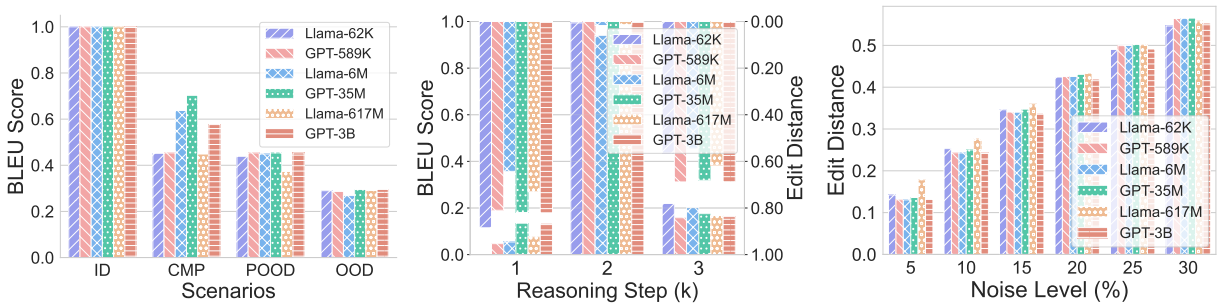


Figure 8 | Task, length, and format generalization of LLMs with different settings. The data distribution lens is invariant across LLMs with various sizes and architectures. Results under more settings are provided in Figure 15.

8.2. External Validity

The key to the external validity of the data distribution lens is to identify the distribution discrepancy between training data and test queries, which makes direct evaluation extremely challenging due to the opacity of the training data used by SOTA LLMs. However, this problem can be alleviated if we can curate data unseen during training and then use it to fine-tune LLMs. By interfering with data generated by DATAALCHEMY, where LLMs produce totally random answers, we confirm the vanity of the proposed pipelines.

Experiment setup. We conduct the experiments of task, length, and format generalization fine-tuning two SOTA LLMs: LLaMA3-8B-Instruct [Dubey et al. \(2024\)](#) and Qwen3-14B [Yang et al. \(2025\)](#).

Findings. As shown in Figure 9, the performance of SOTA LLMs exhibits similar trends to pre-trained models in DATAALCHEMY across task, length, and format generalization, indicating the external validity of the distribution lens. Additional results and analysis are provided in Appendix F.6.

9. Discussion and Implication

Through rigorous experiments, we demonstrate that CoT reasoning is effective when applied to (near) in-distribution data, but becomes fragile and prone to failure even under moderate distribution shifts. What appears to be structured reasoning can be a mirage, emerging from memorized or interpolated

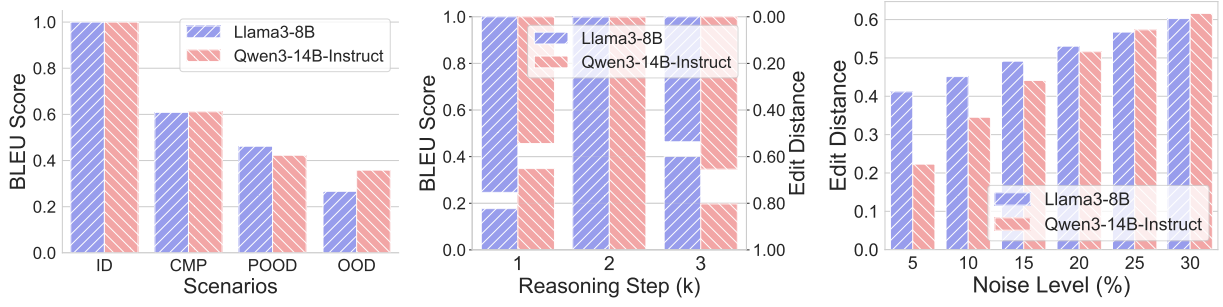


Figure 9 | Task, length, and format generalization of SOTA LLMs. The data-distribution lens is valid.

patterns in the training data rather than logical inference. Our work carries important implications for both LLM researchers and practitioners, which are further discussed in Appendix I.

10. Conclusion

We examine the CoT reasoning of LLMs through the data distribution lens, revealing that the perceived structured reasoning capability largely arises from inductive biases shaped by in-distribution training data, whose effectiveness is bounded by distribution discrepancies. We propose a fully controllable framework, DATAALCHEMY, and systematically probe CoT reasoning with distribution discrepancies introduced by *tasks*, *length* and *format*. Comprehensive experiments confirm that the data distribution is invariant across LLMs with different architectures and sizes. We hope DATAALCHEMY can serve as a platform where researchers can rigorously explore the nature of LLMs, inspiring the discovery of universal principles.

Limitations

While our work offers a rigorous, controlled investigation into the nature of Chain-of-Thought (CoT) reasoning, we acknowledge several limitations that provide avenues for future research:

(i) *Synthetic environment vs. natural language complexity.* Our controlled experiments on the abstract environment DATAALCHEMY, which distills real-world language tasks into symbolic atoms, elements, and transformations. While this abstraction enables full and fine-grained control over distribution factors and avoids data leakage, it may inevitably not fully capture the semantic richness, ambiguity, and compositional diversity present in natural language. While external validity of the proposed data distribution is confirmed by real-world SOTA LLMs, the observed brittleness of CoT reasoning under distribution shifts may manifest more stealthily, sophisticatedly, and task-dependently in more complex real-world settings.

(ii) *Distribution discrepancy measurement and data opacity.* Although we evaluate a wide range of model architectures, sizes, and temperatures, including both models trained from scratch and state-of-the-art pretrained LLMs, the training data distributions of commercial or large proprietary models remain uncovered due to the opaque nature of training data and model weights. As a result, fully estimating the distribution discrepancy between pretraining data and test queries is inherently challenging, limiting the precision with which our data distribution lens can be quantitatively validated in fully realistic and transparent scenarios.

(iii) *Scope of generalization dimensions.* We focused our analysis on three specific dimensions of generalization: task, length, and format. While these cover a broad spectrum of OOD scenarios, we

did not explicitly model other forms of distribution shift, such as cross-lingual transfer, multi-modal reasoning, or shifts in cultural context.

Ethical Considerations

This work studies the reasoning behavior of large language models and does not involve human subjects, personal data, or user-generated content. All experiments are conducted on synthetic or publicly available benchmarks and models, and models trained from scratch use data generated entirely within the proposed framework, avoiding issues of privacy, consent, or data misuse.

Our findings highlight that CoT reasoning can produce fluent yet logically inconsistent or unfaithful reasoning traces when models are evaluated outside their training distributions. This has ethical implications for the deployment of LLMs in high-stakes applications such as education, healthcare, law, and scientific decision-making, where users may over-trust seemingly coherent reasoning explanations. We emphasize that the presence of a detailed reasoning trace should not be equated with correctness, reliability, or genuine understanding.

By exposing the limitations and fragility of CoT reasoning, this work aims to promote more responsible use of LLMs and encourage the research community to develop evaluation protocols and modeling approaches that better reflect true generalization and reasoning capabilities. We believe that transparency about these limitations is essential for preventing misuse and misinterpretation of LLM-generated reasoning.

References

- O. Bentham, N. Stringham, and A. Marasovic. Chain-of-thought unfaithfulness as disguised accuracy. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. Reproducibility Certification.
- M. Budnikov, A. Bykova, and I. P. Yamshchikov. Generalization potential of large language models. *Neural Computing and Applications*, 37(4):1973–1997, 2025.
- Q. Chen, L. Qin, J. Liu, D. Peng, J. Guan, P. Wang, M. Hu, Y. Zhou, T. Gao, and W. Che. Towards reasoning era: A survey of long chain-of-thought for reasoning large language models. *arXiv preprint arXiv:2503.09567*, 2025a.
- Y. Chen, J. Benton, A. Radhakrishnan, J. Uesato, C. Denison, J. Schulman, A. Somani, P. Hase, M. Wagner, F. Roger, et al. Reasoning models don’t always say what they think. *arXiv preprint arXiv:2505.05410*, 2025b.
- H. Cho, J. Cha, P. Awasthi, S. Bhojanapalli, A. Gupta, and C. Yun. Position coupling: Improving length generalization of arithmetic transformers using task structure. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- S. Garg, D. Tsipras, P. S. Liang, and G. Valiant. What can transformers learn in-context? a case study of simple function classes. *Advances in neural information processing systems*, 35:30583–30598, 2022.
- D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.